

CO2_AT1 Scenario Based Question

Question-1:

Define a class named movieMagic with the following description: (20)

Instance variables/data members:

int year – to store the year of release of a movie

String title – to store the title of the movie.

float rating – to store the popularity rating of the movie.

(minimum rating = 0.0 and maximum rating = 5.0)

Member Methods:

(i) void accept() To input and store year, title and rating.

(ii) void display() To display the title of a movie and a message based on the rating as per the table below.

RATING	MESSAGE TO BE DISPLAYED
0.0 to 2.0	Flop
2.1 to 3.4	Semi-hit
3.5 to 4.5	Hit
4.6 to 5.0	Super Hit

Write a main method to create an object of the class and call the above member methods.

```
#include <iostream>
#include <string>
using namespace std;

class movieMagic
{
    int year;
    string title;
    float rating;

public:
    void accept()
    {
        cout << "Enter year: ";
        cin >> year;

        cin.ignore();
        cout << "Enter title: ";
        getline(cin, title);

        cout << "Enter rating: ";
        cin >> rating;
    }

    void display()
    {
        cout << "\nMovie Title: " << title << endl;
```

```

        if (rating >= 0.0 && rating <= 2.0)
            cout << "Flop";
        else if (rating <= 3.4)
            cout << "Semi-hit";
        else if (rating <= 4.5)
            cout << "Hit";
        else if (rating <= 5.0)
            cout << "Super Hit";
        else
            cout << "Invalid rating";
    }
};

int main()
{
    movieMagic m;
    m.accept();
    m.display();

    return 0;
}

```

Question-2:

Define a class named BookFair with the following description: (20)

Instance variables/Data members:

String Bname – stores the name of the book.

double price – stores the price of the book.

Member Methods:

(i) void Input() – To input and store the name and the price of the book.

(ii) void calculate() – To calculate the price after discount. Discount is calculated based on the following criteria.

PRICE	DISCOUNT
Less than or equal to Rs 1000	2% of price
More than Rs 1000 and less than or equal to Rs 3000	10% of price
More than Rs 3000	15% of price

(iv) void display() – To display the name and price of the book after discount.

Write a main method to create an object of the class and call the above member methods

```

#include <iostream>
#include <string>
using namespace std;

```

```

class BookFair
{
    string Bname;

```

```

double price;
double discount;
double finalPrice;

public:
void Input()
{
    cout << "Enter book name: ";
    cin.ignore();
    getline(cin, Bname);

    cout << "Enter price: ";
    cin >> price;
}

void calculate()
{
    if (price <= 1000)
        discount = price * 0.02;
    else if (price <= 3000)
        discount = price * 0.10;
    else
        discount = price * 0.15;

    finalPrice = price - discount;
}

void display()
{
    cout << "\nBook Name: " << Bname << endl;
    cout << "Price after discount: " << finalPrice << endl;
}
};

int main()
{
    BookFair b;
    b.Input();
    b.calculate();
    b.display();

    return 0;
}

```

Question-3:

Define a class Electric Bill with the following specifications:

Instance Variable/ data member:

String n – to store the name of the customer

int units – to store the number of units consumed

double bill – to store the amount to paid

Member methods:

Void accept() – to accept the name of the customer and number of units consumed

Void calculate() – to calculate the bill as per the following tariff :

Number of units Rate per unit

First 100 units Rs.2.00

Next 200 units Rs.3.00

Above 300 units Rs.5.00

A surcharge of 2.5% charged if the number of units consumed is above 300 units.

Void print() – To print the details as follows :

Name of the customer

Number of units consumed

Bill amount

Write a main method to create an object of the class and call the above member methods.

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
class ElectricBill
```

```
{  
    string n;  
    int units;  
    double bill;
```

```
public:
```

```
    void accept()  
    {  
        cout << "Enter customer name: ";  
        cin.ignore();  
        getline(cin, n);  
  
        cout << "Enter units consumed: ";  
        cin >> units;  
    }
```

```
    void calculate()  
    {  
        if (units <= 100)  
        {  
            bill = units * 2.0;  
        }
```

```

        else if (units <= 300)
        {
            bill = 100 * 2.0 + (units - 100) * 3.0;
        }
        else
        {
            bill = 100 * 2.0 + 200 * 3.0 + (units - 300) * 5.0;
            bill = bill + bill * 0.025;
        }
    }

    void print()
    {
        cout << "\nName of the customer: " << n << endl;
        cout << "Number of units consumed: " << units << endl;
        cout << "Bill amount: " << bill << endl;
    }
};

int main()
{
    ElectricBill e;
    e.accept();
    e.calculate();
    e.print();

    return 0;
}

```

Question-4:

Design a class RailwayTicket with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

String coach: to store the type of coach customer wants to travel. long mobno: to store customer's mobile number.

int amt: to store basic amount of ticket.

int totalamt: to store the amount to be paid after updating the original amount.

Methods:

void accept(): to take input for name, coach, mobile number and amount.

void update(): to update the amount as per the coach selected. Extra amount to be added in the amount as follows:

Type of coaches	Amount
First_AC	700
Second_AC	500
Third_AC	250
sleeper	None

void display(): To display all details of a customer such as name, coach, total

amount and mobile number.

Write a main() method to create an object of the class and call the above methods.

```
#include <iostream>
#include <string>
using namespace std;

class RailwayTicket
{
    string name;
    string coach;
    long mobno;
    int amt;
    int totalamt;

public:
    void accept()
    {
        cout << "Enter name: ";
        cin.ignore();
        getline(cin, name);

        cout << "Enter coach: ";
        cin >> coach;

        cout << "Enter mobile number: ";
        cin >> mobno;

        cout << "Enter basic amount: ";
        cin >> amt;
    }

    void update()
    {
        if (coach == "First_AC")
            totalamt = amt + 700;
        else if (coach == "Second_AC")
            totalamt = amt + 500;
        else if (coach == "Third_AC")
            totalamt = amt + 250;
        else if (coach == "sleeper")
            totalamt = amt;
        else
            totalamt = amt;
    }

    void display()
    {
        cout << "\nName: " << name << endl;
        cout << "Coach: " << coach << endl;
```

```

        cout << "Mobile Number: " << mobno << endl;
        cout << "Total Amount: " << totalamt << endl;
    }
};

int main()
{
    RailwayTicket r;
    r.accept();
    r.update();
    r.display();

    return 0;
}

```

Question-5:

Design a class named ShowRoom with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

long mobno: to store the mobile number of the customer.

double cost: to store the cost of the items purchased.

double dis: to store the discount amount.

double amount: to store the amount to be paid after discount.

Member methods:

void input(): to input customer name, mobile number, cost.

void calculate(): to calculate discount on the cost of purchased items, based on the following criteria:

COST DISCOUNT (IN PERCENTAGE)

Less than or equal to Rs. 10000 5%

More than Rs. 10000 and less than or equal to Rs. 20000 10%

More than Rs. 20000 and less than or equal to Rs. 35000 15%

More than Rs. 35000 20%

void display(): to display customer name, mobile number, amount to be paid after discount.

Write a main() method to create an object of the class and call the above member methods.

```

#include <iostream>
#include <string>
using namespace std;

```

```

class ShowRoom

```

```

{
    string name;
    long mobno;
    double cost;
    double dis;
    double amount;

```

```

public:

```

```

void input()
{
    cout << "Enter customer name: ";
    cin.ignore();
    getline(cin, name);

    cout << "Enter mobile number: ";
    cin >> mobno;

    cout << "Enter cost: ";
    cin >> cost;
}

void calculate()
{
    if (cost <= 10000)
        dis = cost * 0.05;
    else if (cost <= 20000)
        dis = cost * 0.10;
    else if (cost <= 35000)
        dis = cost * 0.15;
    else
        dis = cost * 0.20;

    amount = cost - dis;
}

void display()
{
    cout << "\nCustomer Name: " << name << endl;
    cout << "Mobile Number: " << mobno << endl;
    cout << "Amount to be paid: " << amount << endl;
}
};

int main()
{
    ShowRoom s;
    s.input();
    s.calculate();
    s.display();

    return 0;
}

```